

Nidhi Gohel

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EDUCATION

York University

Bachelor of Science in Applied Mathematics

Toronto, ON

Expected December 2026

- **Relevant Coursework:** Probability & Statistics, Stochastic Processes, Linear Algebra, Calculus, Mathematical Proofs, Logic, Data Structures & Algorithms

QUANTITATIVE PROJECTS

Kalshi Prediction-Market Mispricing Engine | Python, team of 2, in progress

Sep 2026 – Present

- Co-building an engine that scans live Kalshi prediction markets for pricing inconsistencies and paper-trades flagged opportunities against live order-book data, with no real capital at risk.
- Own the mathematical layer: encoding no-arbitrage relationships between linked contracts (mutually exclusive brackets must price to \$1; nested thresholds must be monotone) and detecting violations through structural checks and linear-programming feasibility tests.
- Designing Kelly-criterion position sizing and a signal-scoring model that ranks opportunities by fee-adjusted expected value, so only edges that survive Kalshi's transaction costs reach the paper-trading book.
- Integrating signals with an execution-cost model and backtester to compare expected edge against realized paper-trading P&L.

Market Sentiment Analyzer | Python, HuggingFace Transformers (FinBERT), Scikit-learn

May 2025 – Jun 2025

- Achieved a 12% F1-score improvement over baseline (0.71 to 0.80) by fine-tuning FinBERT on a 5,000-headline domain-specific financial news corpus, outperforming a TF-IDF + Logistic Regression baseline on held-out test data.
- Reduced preprocessing latency by ~40% by implementing batched tokenization and text-cleaning pipelines in Python, enabling scalable inference across 5,000+ unstructured headlines.
- Ensured reproducible model evaluation across 3 transformer configurations by tracking precision, recall, and F1 on stratified validation sets, surfacing the optimal checkpoint for deployment.

Media Consumption & Engagement Analytics Tool | Python, Pandas, NumPy

Dec 2025 – Present

- Surfaced 4 statistically significant engagement patterns across 8,000+ structured records through frequency, duration, and time-based usage analysis, providing insights unavailable in the raw dataset.
- Achieved 100% pipeline reproducibility across 3 analyst runs by standardizing data-cleaning and transformation steps in modular Python scripts, eliminating ad-hoc fixes that previously caused inconsistencies.
- Reduced future onboarding time by an estimated 50% by documenting all data-logic and processing decisions in a structured README, enabling a new contributor to reproduce results without tribal knowledge.

Executive Performance & KPI Analytics Dashboard | SQL, Power BI, DAX

Apr 2025 – May 2025

- Reduced ad-hoc reporting requests by ~60% by translating 10+ unstructured business questions into a self-serve Power BI dashboard with clearly defined KPIs consumed by 3 simulated executive personas.
- Improved trend and anomaly detection by designing a star-schema SQL model (2 fact tables, 5 dimension tables) and implementing MoM/YoY DAX variance calculations, automatically flagging performance shifts of 5% or more.
- Cut dashboard load time to under 3 seconds by optimizing DAX measures and pre-aggregating slow dimensions in SQL views, directly improving executive usability and adoption.

Trading Buddy – Stock Market Web Platform | Node.js, React.js, REST APIs

Jan 2024 – Apr 2024

- Delivered real-time market data for 50+ ticker symbols by integrating 2 third-party financial REST APIs into a Node.js backend, achieving <500 ms average response latency under local load tests.
- Eliminated unauthorized session access by implementing JWT authentication with token expiry and refresh logic, securing identity management across multi-page navigation for all users.
- Increased user engagement surface by 3x by architecting a modular UI with discussion threads and interactive financial guides, enabling future feature expansion without code rewrites.

EXPERIENCE

IKEA

Jun 2023 – Mar 2026

Returns & Exchanges Co-Worker (Data & Operations)

Toronto, ON

- Reduced manual reconciliation time by 12% (~4 hrs/week to ~3.5 hrs/week) by automating 5+ returns-tracking and reporting workflows using Excel macros, SharePoint lists, and MS Office 365 integrations.
- Improved POS system uptime by 18% during peak return windows by systematically diagnosing and escalating incidents to IT, cutting average incident-to-resolution time from 55 min to 45 min.
- Accelerated data-driven decisions for 3 department managers by building Power BI dashboards that surfaced real-time return patterns, defect categories, and customer resolution metrics, replacing a weekly manual report.

TECHNICAL SKILLS

Quantitative: Probability, Stochastic Processes, Expected Value & Kelly Sizing, Linear Programming, Statistical Testing

Languages & Databases: Python, SQL, Java, JavaScript; PostgreSQL, MySQL, Star-Schema Modeling

ML & Data: Pandas, NumPy, Scikit-learn, Matplotlib, LangChain, HuggingFace Transformers (FinBERT)

Analytics: KPI/KRI Design, Trend & Variance Analysis, A/B Testing Concepts, Data Cleaning, Scenario Analysis

Web & Tooling: React.js, Node.js, REST APIs, HTML, CSS, jQuery; Power BI, DAX, Git, GitHub, Jupyter, VS Code, MS Office 365, SharePoint